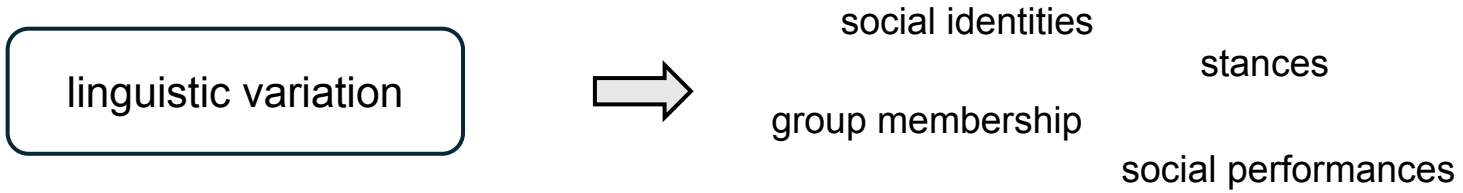
Effects of CV coarticulation on social evaluations of fronted /s/

Grace Brown, Meghan Sumner, Rob Podesva | *Stanford University*

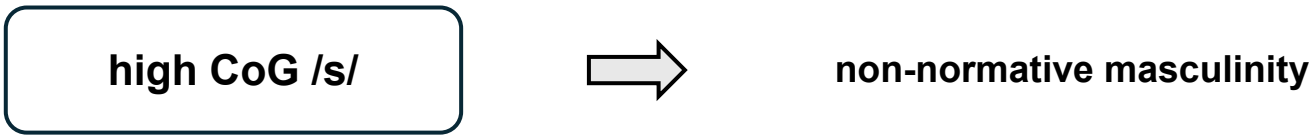
LSA 2026
January 11, 2026

Variation facilitates perception of social meaning

- Perceptual sociolinguistics: listeners have structured knowledge about the linguistic patterns of groups of speakers and individuals



- We map social meanings onto particular linguistic forms:



How does contextual variation contribute to our understanding of social meaning mappings?

Variation is central to speech perception

- Listeners are highly sensitive to variation both **within** and **across** speakers:
  - process upcoming words
  - craft detailed acoustic memories
  - adapt to new speakers
  - use speaker info in speaker recognition, perception, and comprehension

- Sociolinguistic approaches to investigating perceived social meaning have not yet incorporated this finding

phonological variation

phonetic detail

variation in context

e.g., Beddor et al., 2013; Johnson, 1997; Palmeri, Goldigner, & Pisoni, 1993; Bradlow & Bent, 2008; Bradlow & Pisoni, 1999

Revisiting /s/

Gender and /s/:

- /s/ acoustics gender identity and performance
  - High center of gravity /s/ female voices, femininity, and non-normative masculinity
    - (e.g., Strand & Johnson, 2006; Munson, 2007)

Matched guise and perceiving social meaning:

- English listeners hold consistent social evaluation of fronted (high CoG) /s/ across:
  - **speakers** (Campbell-Kibler, 2011)
  - **language context** (Boyd et al., 2021)

Tendency to use speakers whose voices are consistent with normative expectations of gender

Tendency to use invariant /s/ tokens across different phonological environments within a given guise

Representative investigation of /s/ and masculinity



”backed”
CoG: 5600 Hz



“mid”
CoG: 6300 Hz



“fronted”
CoG: 7800 Hz



“Very masculine”

“Very masculine”

“Somewhat masculine”

“Somewhat masculine”

“Neither masculine nor non-masculine”

“Somewhat masculine”

“Somewhat non-masculine”

“Somewhat non-masculine”

“Very non-masculine”

Representative investigation of /s/ and masculinity

At the pie
took a cl
photo of
[s]ea.

”back
CoG: 5

he pier, I
k a clear
oto of the
[s]ea.

“fronted”
CoG: 7500 Hz

“High CoG (fronted) /s/ is perceived as less masculine than low CoG (backed) /s/”

(Campbell-Kibler, 2011; Boyd et al., 2021)



“Very masculine”

“Very masculine”

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“Somewhat masculine”

“Neither masculine nor non-masculine”

“Somewhat masculine”

“Somewhat non-masculine”

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Variation in context

Listeners evaluate the social meaning of linguistic variables **in context:**

- Surrounding phonetic or phonological environment
- Speaker(s)
- Utterance content
- Discourse context
- Social context
- …

Eckert, 2008; Silverstein, 2003

Variation in context

Listeners evaluate the social meaning of linguistic variables **in context**

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- **Speaker(s)**
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Eckert, 2008; Silverstein, 2003

Variation in context

Listeners evaluate the social meaning of linguistic variables **in context**

- **Surrounding phonetic or phonological environment**

Adjacent vowels influence /s/ production

Soli 1981

Variation in context

Listeners evaluate the social meaning of linguistic variables **in context**

- **Surrounding phonetic or phonological environment**

  Adjacent vowels influence /s/ production

- **Speaker(s)**

  Speakers have different phonetic footprints

How do variable speaker and phonetic environments influence social evaluations of /s/?

How do stable vs variable /s/ variants impact these processes?

Matched Guise Design

Between-subjects conditions:
Single-token (n = 270) vs multi-token (n = 269)

Within-subjects conditions (2x3 design):

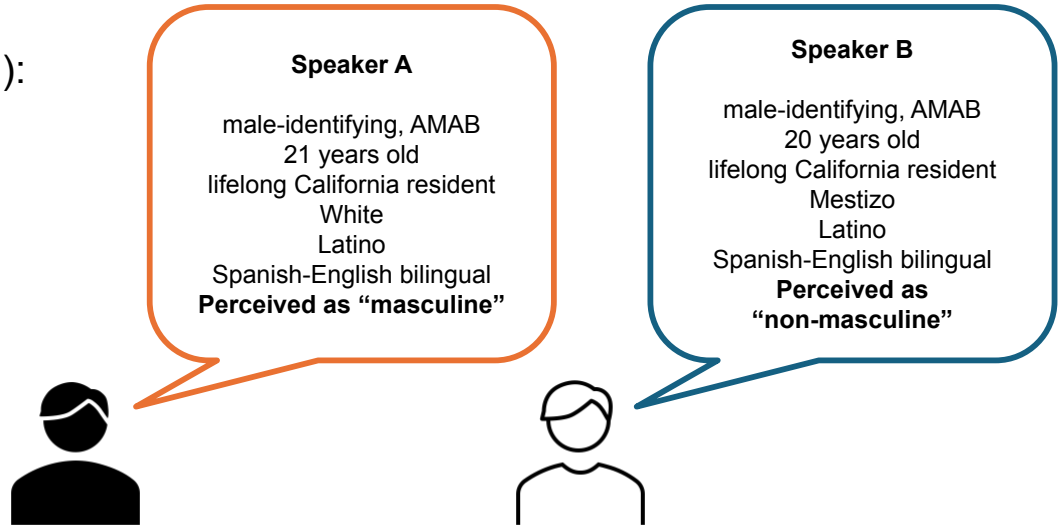
- Speaker: 2 speakers

Matched Guise Design

Between-subjects conditions:
Single-token (n = 270) vs multi-token (n = 269)

Within-subjects conditions (2x3 design):

- Speaker: 2 speakers



Matched Guise Design

Between-subjects conditions:
Single-token (n = 270) vs multi-token (n = 269)

Within-subjects conditions (2x3 design):

- Speaker: 2 speakers
- /s/ conditions:
  - **Front /s/**
  - **Mid /s/**
  - **Back /s/**

Participants heard multiple combinations of each speaker : /s/ condition (n = 72)

Matched Guise Design

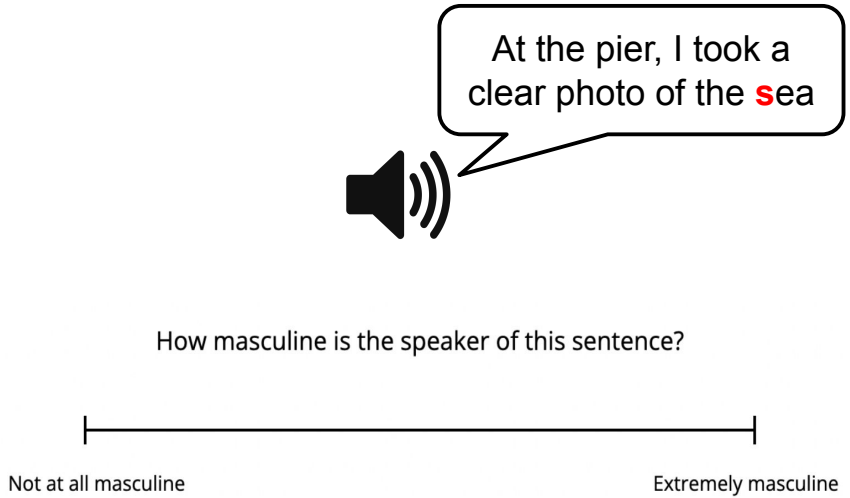
Between-subjects conditions:
Single-token (n = 270) vs multi-token (n = 269)

Within-subjects conditions (2x3 design):

- Speaker: 2 speakers
- /s/ conditions:
  - **Front /s/**
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  - **Back /s/**

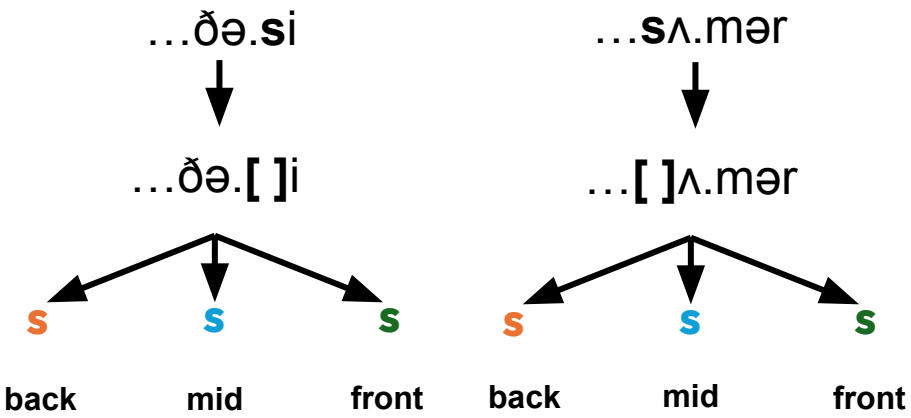
Participants heard multiple combinations of each speaker : /s/ condition (n = 72)

All critical trials had a single, word-initial, prevocalic /s/ token



Single vs multi /s/ conditions

Single-token splicing condition



→ One /s/ token per articulatory guise (realized as CoG):

- Front: 8203 Hz
- Mid: 6928 Hz
- Back: 5654 Hz

→ Spliced tokens matched original for intensity and duration

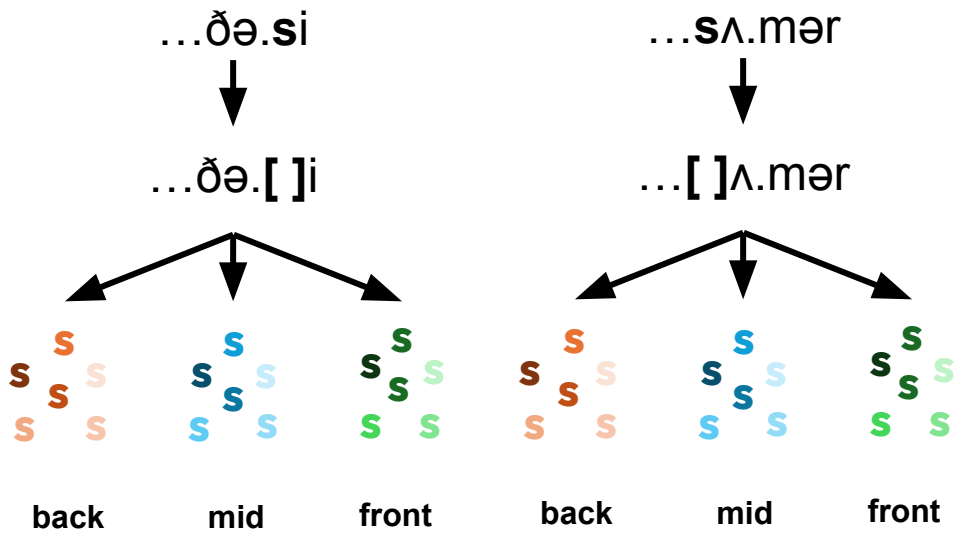
Single vs multi /s/ conditions

→ 18 /s/ tokens per /s/ guise (realized as CoG):

- Mean Front: 8033 Hz
- Mean Mid: 6502 Hz
- Mean Back: 5605 Hz

→ One token per unique phonetic environment

Multi-token splicing condition



Single vs multi /s/ conditions

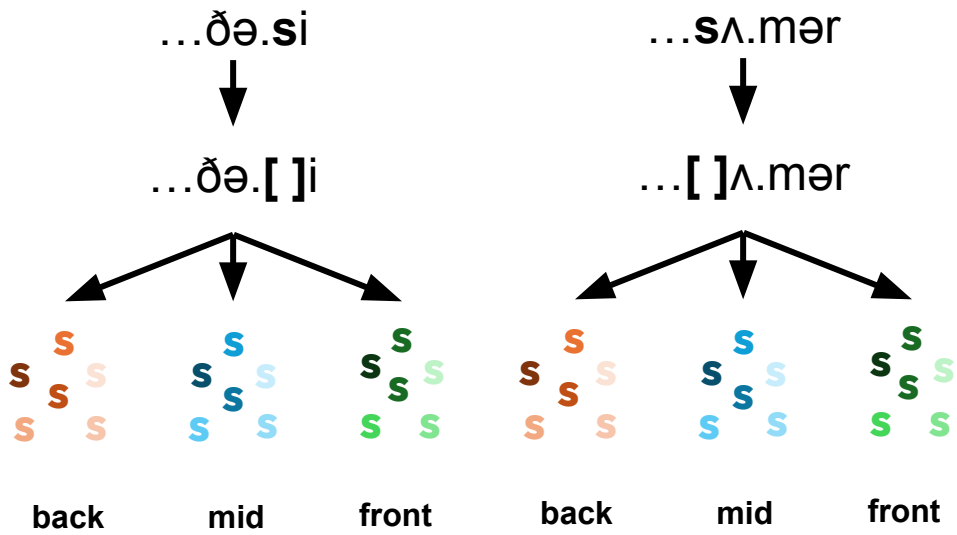
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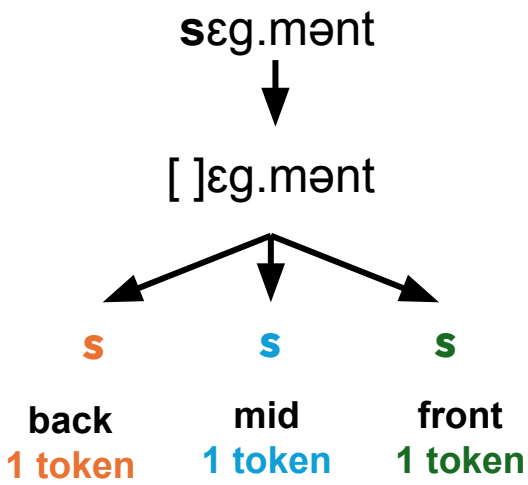
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Multi-token splicing condition

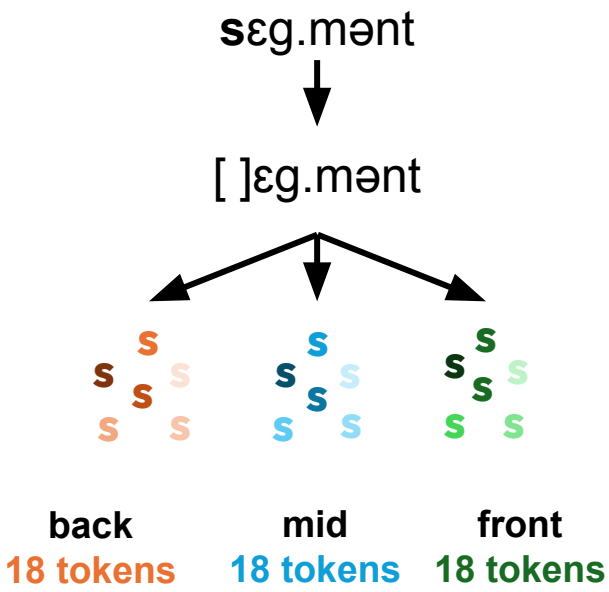


Single vs multi /s/ conditions

Single-token splicing



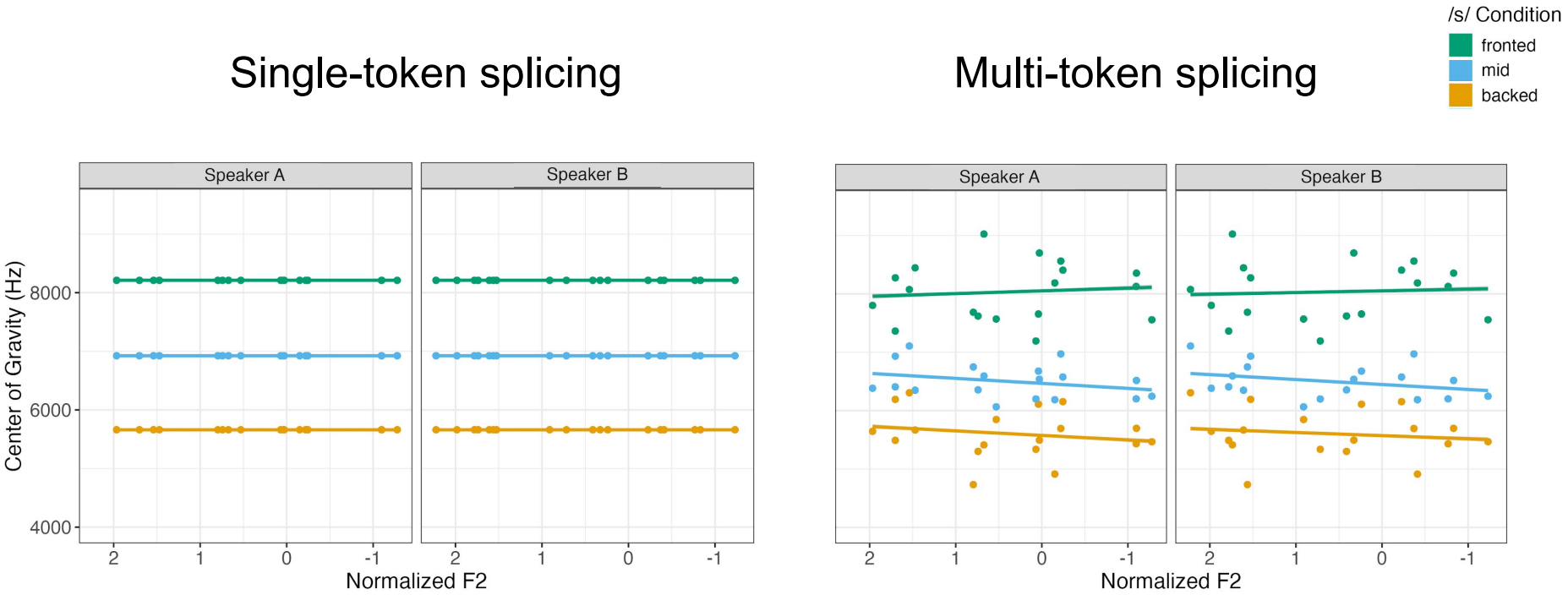
Multi-token splicing



Single vs multi /s/ conditions

Single-token splicing

Multi-token splicing



Statistical Modeling

Zero-one inflated beta regression:

Masculinity rating (mu) ~ splicing design * speaker * /s/ condition * raw F2 * /s/ duration (ms) + proportion of creak + mean F0 + F0 range + (1 | participant) + (1 | item)

Response variability:

phi ~ speaker

Response scale endpoint usage:

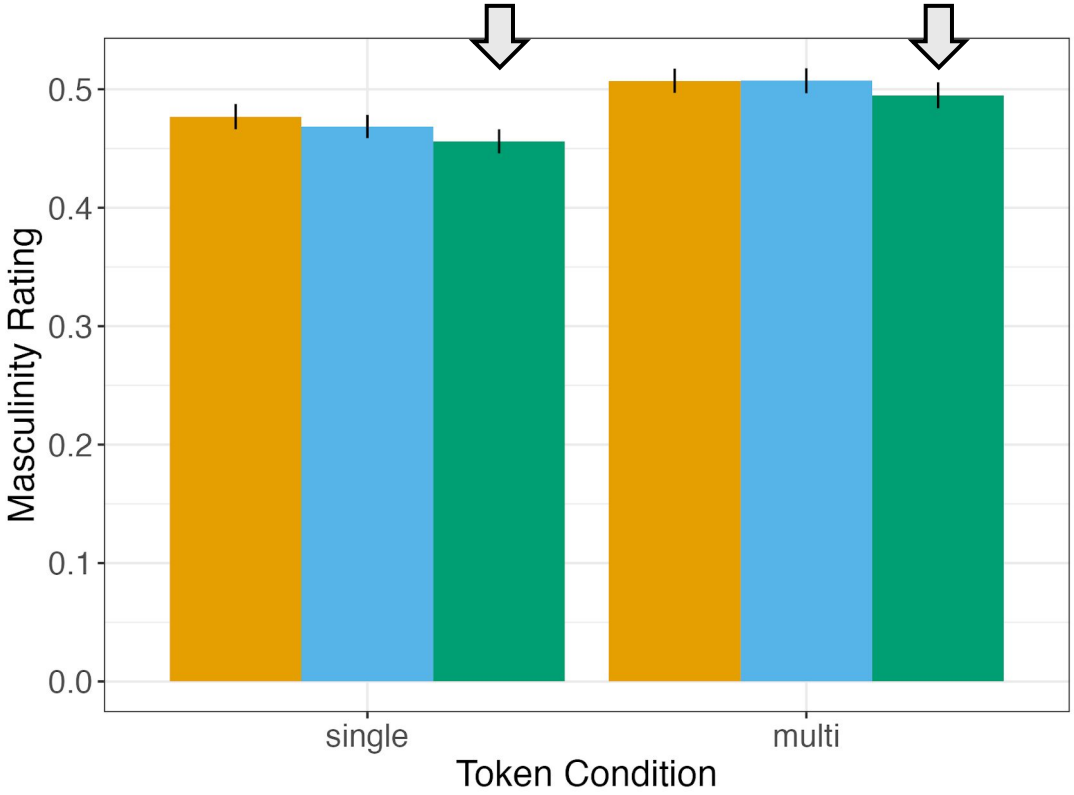
Zero-one inflation (zoi) ~ speaker

Conditional one inflation (coi) ~ speaker

Q1: On average, how do listeners evaluate the masculinity of fronted /s/?

Does this vary by token condition?

Q1 Mean masculinity rating by /s/ condition and token condition



Front /s/ is perceived as less masculine than mid /s/ and back /s/

This pattern is consistent across token conditions

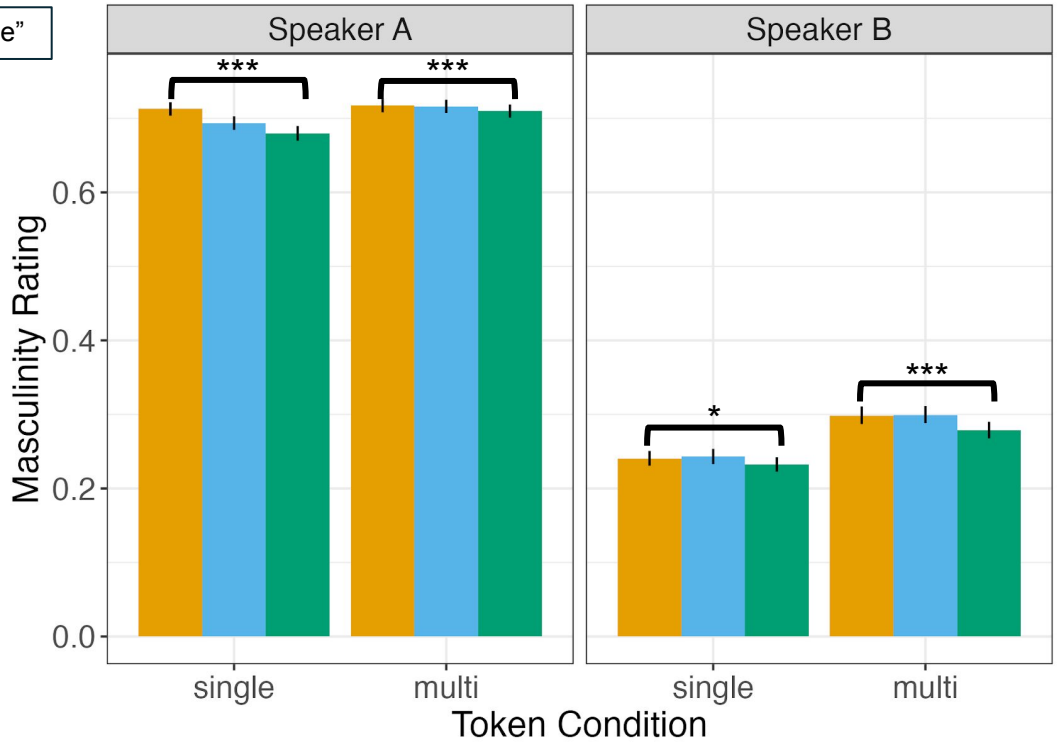
Replication of previous results
(Campbell-Kibler, 2011 & Boyd et al., 2021)

Q2: How does speaker voice impact how listeners evaluate the masculinity of fronted /s/?

Does this vary by token condition?

Q2 Mean masculinity rating by /s/ condition, speaker, and token condition

“more masculine”



“less masculine”

Front /s/ is perceived as less masculine than back /s/

This pattern is relatively consistent across token and speaker conditions

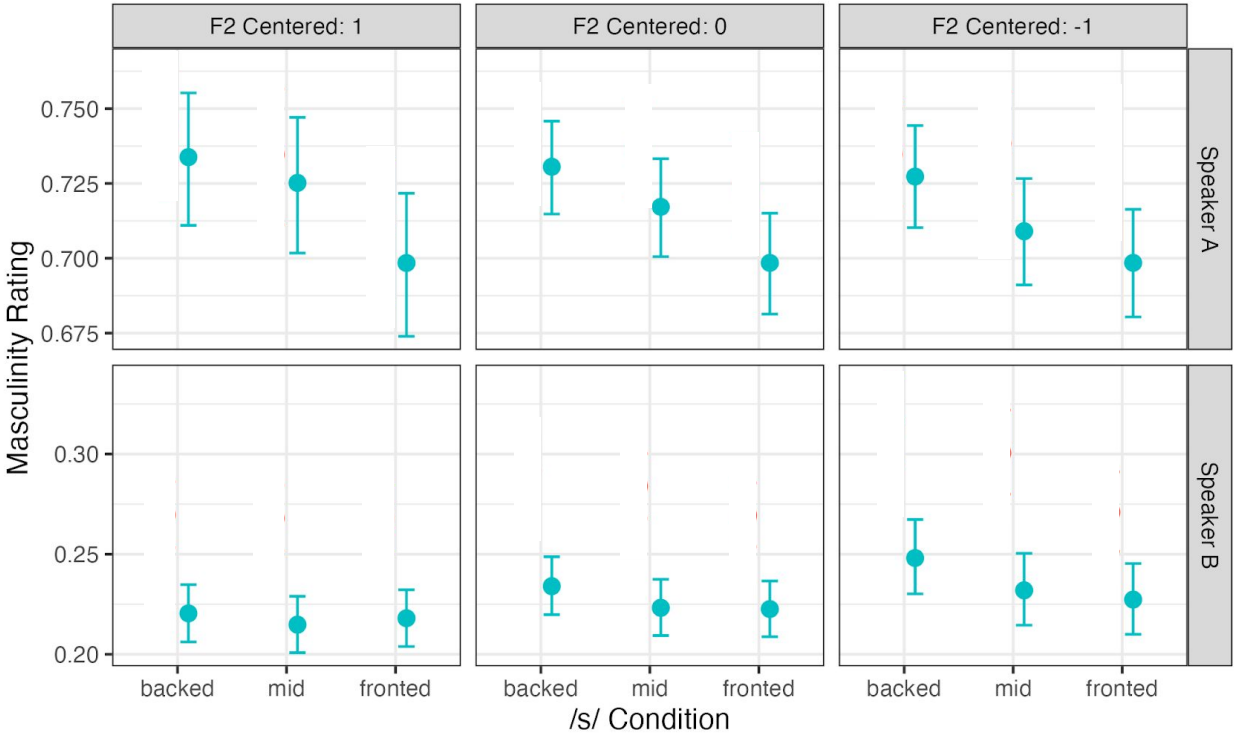
Speaker voice is a greater cue to masculinity rating than /s/

Q3: How do the speaker and the F2 of the following vowel jointly impact how listeners evaluate the masculinity of fronted /s/?

Does this vary by token condition?

Q3

F2 (vowel frontness) increases



Token Condition

multi

single

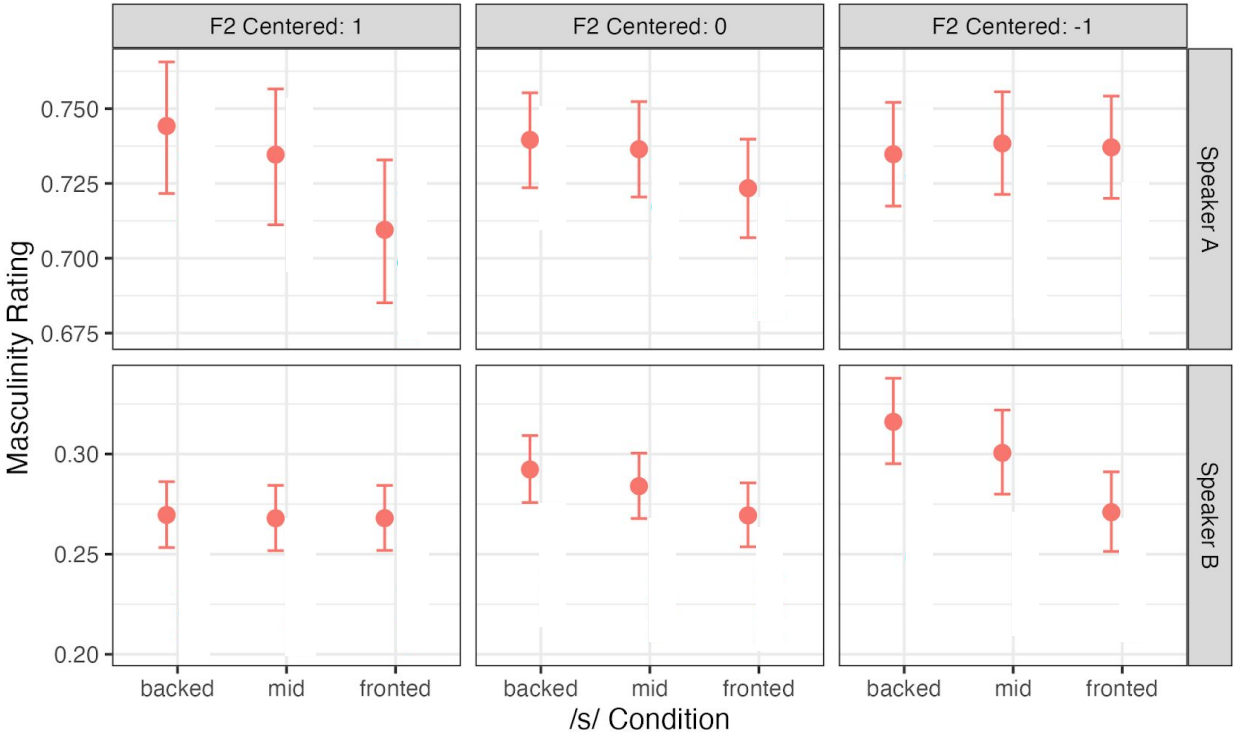
SINGLE TOKEN CONDITION

Listeners perceive differences in masculinity across /s/ conditions in all F2 contexts

Listeners perceive fewer difference in masculinity across /s/ conditions in all F2 contexts

Q3

F2 (vowel frontness) increases



Token Condition

multi

single

MULTI TOKEN CONDITION

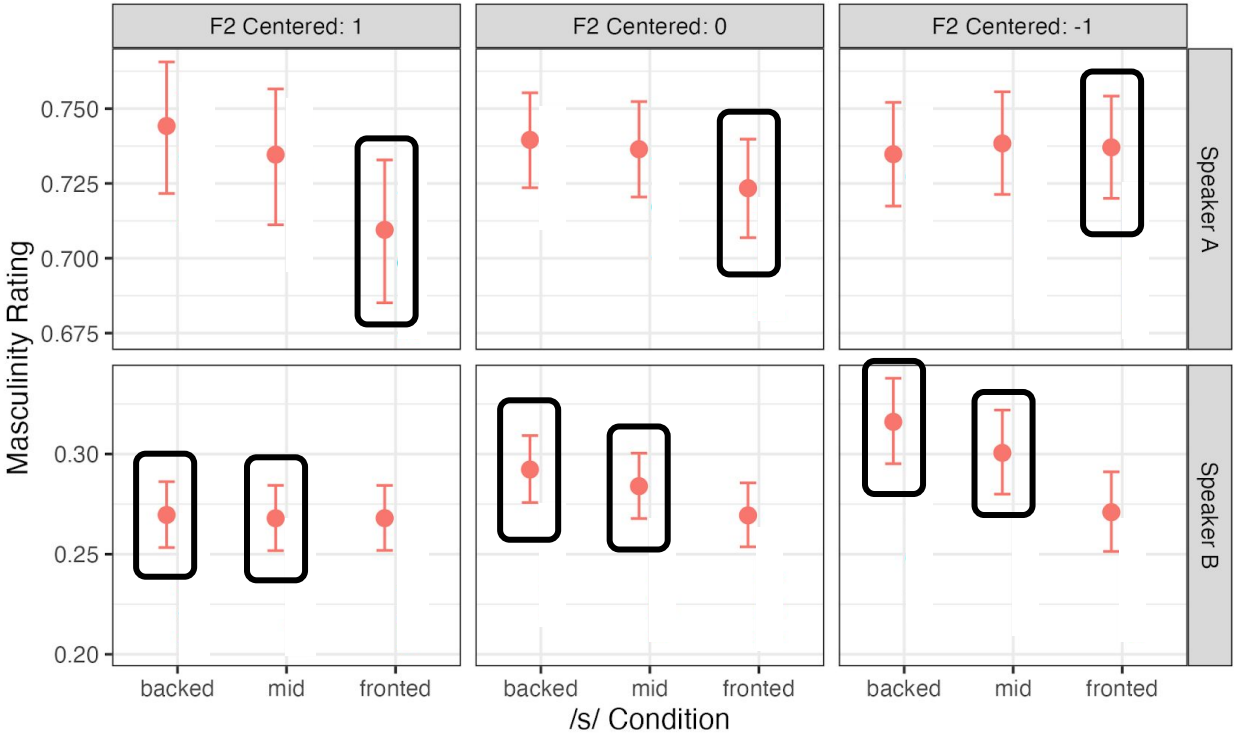
Listeners perceive greater differences in masculinity across /s/ conditions in **high** F2 contexts

Listeners perceive greater differences in masculinity across /s/ conditions in **low** F2 contexts

Q3

F2 (vowel frontness) increases

MULTI TOKEN CONDITION



As F2 ↑:
Fronted /s/ is rated as less masculine
95% CI: [-0.029, -0.002]

As F2 ↑:
Backed /s/ is rated as more masculine
95% CI: [-0.035, -0.016]
Mid /s/ is rated as more masculine
95% CI: [-0.028, -0.008]

How are listeners evaluating /s/ variation in context?

- Generally, **fronted** (high CoG) /s/ is perceived as **less masculine** than backed (low CoG) /s/
- When we consider the combined influence of /s/, F2 of the following vowel, speaker, and splicing technique, the picture changes:

Single Token

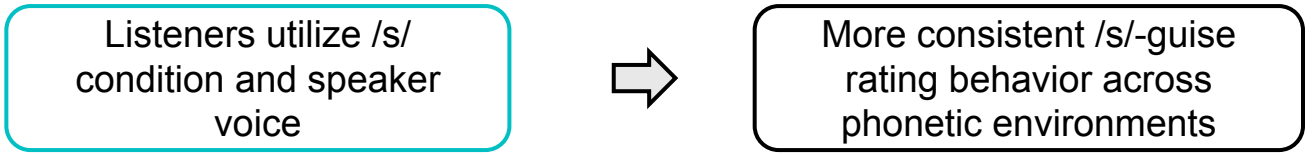
/s/-guise effects on perceived masculinity are mediated by speaker voice

Multi Token

/s/-guise effects on perceived masculinity are jointly mediated by speaker voice and F2 of the following vowel

How are listeners evaluating /s/ variation in context?

- Listeners are always socially evaluating /s/ in context, not as an isolated variable
- When exposed to stable /s/ variants and variable speaker voices:



How are listeners evaluating /s/ variation in context?

- Listeners are always socially evaluating /s/ in context, not as an isolated variable
- When exposed to stable /s/ variants and variable speaker voices:

Listeners utilize /s/ condition and speaker voice → More consistent /s/-guise rating behavior across phonetic environments

- When exposed to variable /s/ variants and variable speaker voices:

Listeners utilize phonetic detail in and around /s/ and speaker voice → More variable /s/-guise rating behavior across speakers and phonetic environments

Moving forward: incorporating greater variation in sociolinguistic perception

This study

multi-token splicing and use of diverse speakers

how listeners use variation **beyond categorical /s/ guises** to inform their social evaluations of speakers

Broader applications

greater variability in sociolinguistic perception stimuli

improved understanding of how listeners use variation to make social inferences in diverse contexts

Thank You!

Special thanks to everyone at the Stanford Phonetics Lab and P-Interest for their feedback and suggestions!

contact: grcbrown@stanford.edu